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Determination of Human Effective Dose via Translational Ocular Modeling: Clinical Translation of a Novel Topical Agent Using PBPK

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ABSTRACT

Wet age-related macular degeneration (wAMD) is a leading cause of vision loss, and human dose selection for ocular therapeutics remains a key translational challenge. We developed a physiologically based pharmacokinetic (PBPK) model for EB-203 to guide dose selection by translating preclinical ocular tissue exposure to human. PK data were obtained from mouse and rabbit studies and human plasma PK from a phase I trial. The model incorporated species-specific ocular physiology, drug permeability, and pharmacokinetics, and was implemented using an open-source PBPK modeling software (MoBi, version 12.2, Open Systems Pharmacology). Drug-specific model parameters were estimated using mouse and rabbit plasma and ocular tissue concentration data. Human plasma PK data from a Phase I clinical trial were then used to estimate human-specific systemic clearance, and human ocular tissue concentrations were subsequently predicted by the model. Simulated human retinal drug concentrations (AUC, $C_{\max}$, and C_{avr}) under multiple dosing regimens were evaluated against preclinical efficacy thresholds to determine the optimal dosing strategy. The model adequately described drug disposition across species, with most predicted PK parameters falling within a 2-fold error of the observed data. Simulations suggested that a 4% four-times-daily dosing regimen in humans achieves retinal exposure comparable to the efficacious mouse concentrations. No significant systemic accumulation was observed, indicating a favorable safety profile. This PBPK framework provides a practical approach for predicting human ocular drug exposure and can be adapted to other compounds or ocular indications in the absence of direct tissue sampling. These results support further clinical evaluation of EB-203 for the treatment of wAMD.

Abbreviations: AMD, age-related macular degeneration; AUC, area under the curve; C_{avr} , average concentration; $C_{\max}$, maximum concentration; CNV, choroidal neovascularization; dAMD, dry age-related macular degeneration; GMFE, geometric mean fold error; IV, intravenous; PBPK, physiologically based pharmacokinetic; PK, pharmacokinetic; $T_{\max}$, time to peak drug concentration; VEGF, vascular endothelial growth factor; wAMD, wet age-related macular degeneration.

Hwi-yeol Yun, Soyoung Lee, Jaewook Yang and Jung-woo Chae contributed equally to this work.

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Study Highlights

- What is the current knowledge on the topic?
 - Human dose selection for topical ophthalmic drugs is difficult because ocular tissue concentrations cannot be sampled directly in humans.
- What question did this study address?
 - Can a PBPK model translate preclinical ocular exposure and human plasma data into human retinal exposure to guide dosing of the topical agent EB-203?
- What does this study add to our knowledge?
 - A cross-species ocular PBPK model predicted human retinal exposure without direct tissue sampling, indicating that 4% EB-203 four times daily achieves exposure comparable to efficacious preclinical levels.
- How might this change clinical pharmacology or translational science?
 - It offers a practical way to predict human ocular exposure and select doses when target-tissue sampling is not feasible, adaptable to other compounds and ocular indications.

1 | Introduction

Age-related macular degeneration (AMD) is a leading cause of blindness in individuals over 50 years of age worldwide [1, 2]. AMD is classified as dry AMD (dAMD) or wet AMD (wAMD), with dAMD potentially progressing to wAMD, which is characterized by choroidal neovascularization and subsequent vision loss [3].

The therapeutic management of wet AMD currently depends on intravitreal anti-VEGF injections [4–7], highlighting the need for alternative delivery strategies that reduce treatment burden. EB-203 is a peptide-based investigational compound (MW 714.77 g/mol) currently in clinical development for wAMD, having completed a Phase I trial (NCT05538949) with Phase II evaluation planned. It exerts its anti-angiogenic effect by binding to ITGB3BP (Integrin beta-3 binding protein), stabilizing the NF- κ B/I κ B α complex and thereby suppressing downstream expression of HIF-1 α , VEGF, SRC, and MMP to inhibit choroidal neovascularization.

Given its peptide backbone and hydrophilic physicochemical profile, CYP-mediated metabolism is not expected to be a primary elimination pathway; proteolytic degradation by endogenous peptidases is anticipated to predominate. Accordingly, the PBPK model was constructed assuming passive diffusion as the sole transport mechanism, with active transport and metabolic clearance not incorporated.

In a laser-induced CNV mouse model, EB-203 inhibited choroidal neovascularization at concentrations of 2%–4% under multiple dosing regimens. In the Phase I clinical trial (NCT05538949), no dose-limiting toxicities were observed across the tested dose range (1%–4%), and treatment-emergent adverse events were mild and self-resolving, with no significant difference in

incidence between the EB-203 and placebo groups. Systemic exposure was minimal across all dose levels.

Despite these encouraging preclinical and early clinical findings, PK-based assessment of efficacy in humans remains challenging, as continuous intraocular tissue PK sampling is not feasible in humans [8].

To address these limitations, a physiologically based pharmacokinetic (PBPK) model was used to translate tissue concentrations associated with efficacy in animal models to humans [9]. By integrating species-specific physiology and leveraging human plasma PK as a clinical anchor, this approach enables cross-species inference of human ocular tissue exposure and supports mechanistic translation of preclinical efficacy.

Given that the fundamental anatomy and physiology of the eye are highly conserved among vertebrates [10, 11], this approach assumes that key structural and functional similarities allow for reasonable interspecies scaling. To maximize the predictive accuracy, we performed model fitting using plasma and ocular tissue concentrations from multiple species. Mouse data were used as a reference for drug efficacy, whereas rabbit ocular data provided structural insights owing to their anatomical similarity to the human eye.

Building on this framework, an integrative PBPK model for EB-203 was constructed to integrate and interpret multi-species data, thereby enabling evaluation of whether efficacious exposure levels can be achieved in humans.

2 | Methods

2.1 | Pharmacokinetic Settings and Exploratory Data Analysis

Pharmacokinetic profiles were obtained from mice, rabbits, and humans. The mouse PK study included concentrations in the plasma, cornea, choroidal area (choroid and sclera), and retina. The rabbit study included concentrations in plasma and multiple ocular tissues, including the aqueous humor, vitreous humor, cornea, conjunctiva, sclera, iris–ciliary body, retina, and choroid (IACUC No. IJUBPH_2022-017-01, IJUBPH-2021-008-03, and IJUBPH_2021-003-06). The human PK profile included plasma concentrations (NCT05538949).

Eight sets of data were used for modeling (design features are summarized in Table S1, and bioanalytical method in Table S2). In the human study, the drug was administered to one eye, whereas in the animal experiments, it was administered to both eyes. In each tissue and dataset, area under the curve of drug concentration (AUC) data were calculated using non-compartmental analysis to identify PK trends and relationships between tissues.

2.2 | Model Parameter Settings and Collection

In addition to PK data, physiological information on ocular tissues of mice, rabbits, and humans was collected from the

literature. The model parameters were divided into two categories: those representing physicochemical properties and those representing the physiological spaces of the drug within the body. The PBPK model was constructed by assuming only passive transport (diffusion) for drug disposition, with no active transport or metabolic mechanisms considered.

Physicochemical properties were estimated using molecular descriptors derived from the drug structure (SMILES; Simplified Molecular Input Line Entry System, lipophilicity and solubility with ALOGPS 2.1 [12]). These values were set as initial estimates for model parameter optimization.

Ocular physiological parameters were defined for the aqueous humor, vitreous humor, cornea, conjunctiva, sclera, iris-ciliary body, retina, and choroid. Tissue volumes were assigned based on sampled tissue mass, assuming a density of 1g/cm³. When species-specific values were unavailable, parameters were extrapolated using allometric scaling based on available cross-species data.

Allometric scaling for ocular parameter dimensions was calculated following the method described by Goldman et al. and West et al. [13, 14], based on the structural similarity of the ocular anatomy between species [10]. When species-specific values were unavailable, the missing values were estimated by single-point allometric scaling from the rabbit, applying the allometric exponent to the eye-volume ratio (Details in Figure S1). The key relationship for allometric scaling is as follows:

$$\text{Flow} \propto \text{Volume}^{0.75}$$

The physiological parameters were scaled according to the eye volume of each species, and key ocular flows from the aqueous humor, trabecular meshwork, and uveoscleral pathway were estimated.

2.3 | Model Structurization

The eye compartment was configured to follow the 4-compartment system provided as a standard in PK-Sim, which includes compartments for blood cells, plasma, and interstitial and intracellular spaces within the eye. Fourteen compartments were incorporated to represent the ocular space (Table S3). Blood cells and plasma compartments were constructed using the standard PK-Sim compartment logic. These compartments are connected to the corresponding compartments in the arterial and venous systems, allowing drug molecules to circulate throughout the body.

In human studies, the drug was administered to only one eye, whereas in animal studies, it was administered to both eyes. To account for this difference, parameters representing the eye compartment and administration were doubled in the model. Given the small size of the untreated eye in humans, any potential diffusion of the drug to the contralateral eye is considered minimal and unlikely to significantly contribute to systemic drug concentration.

2.4 | Drug Transportation Mechanism in Model

Drug transfer between ocular compartments and the systemic circulation was modeled using permeability-limited diffusion and physiological flows, with permeability values calculated using PK-Sim default algorithms. Transport processes were implemented across five interface types: lumen-lumen, lumen-intracellular, intracellular-intracellular, interstitial-intracellular, and interstitial-interstitial. Depending on anatomical connectivity, transport was modeled as either unidirectional (e.g., drainage and absorption pathways) or bidirectional diffusion between adjacent compartments. In addition, aqueous humor flow, trabecular meshwork flow, and uveoscleral flow were incorporated.

Lumen-lumen transport, representing drug movement from the precorneal area to the nasolacrimal duct and subsequently to the stomach, was described as follows:

$$\frac{dA_{\text{drug}}}{dt} = w * A_{\text{drug}}$$

Lumen-intracellular transport, including pathways from the precorneal area to the cornea and conjunctiva and from the nasolacrimal duct to plasma, was described as follows:

$$\frac{dA_{\text{drug}}}{dt} = \text{Surface area} \times (\text{Permeability} \times P_{\text{factor}}) \times C_{\text{lumen}}$$

Intracellular-intracellular transport between ocular tissues (conjunctiva, sclera, choroid, retina, iris-ciliary body, cornea, aqueous humor, and vitreous humor) and interstitial-interstitial transport were described as:

$$\frac{dA_{\text{drug}}}{dt} = \text{Surface area} \times (\text{Permeability} \times P_{\text{factor}}) \times (C_{\text{source}} - C_{\text{target}})$$

Interstitial-intracellular transport followed the PK-Sim default formulation:

$$\frac{dA_{\text{drug}}}{dt} = \text{Surface area}_{\text{int-cell}} \times \text{Permeability} (K_{\text{water/int}} \times P_{\text{int-cell}} \times C_{\text{int}} - K_{\text{water/cell}} \times P_{\text{cell-int}} \times C_{\text{cell}})$$

The interstitial-intracellular contact surface area was estimated from compartment volume assuming spherical geometry, with a correction factor estimated during model fitting:

$$\text{Surface area}_{\text{unit}} = 4\pi \times V_{\text{compartment}}^{0.66}$$

$$\text{Surface}_{\text{int-cell}} = \text{SA}_{\text{factor}} \times \text{Surface area}_{\text{unit}}$$

$$\frac{dA_{\text{drug}}}{dt} = \text{Surface area} \times (\text{Permeability} \times P_{\text{factor}}) \times (C_{\text{source}} - C_{\text{target}})$$

Ocular fluid transport, including aqueous humor, trabecular meshwork, and uveoscleral flows, was described as follows:

$$\frac{dA_{\text{drug}}}{dt} = \text{Weight}_{\text{tissue}} \times Q_{\text{flow}} \times C_{\text{tissue}}$$

Aqueous humor flow was modeled from the vitreous humor to the aqueous humor, trabecular meshwork flow from the aqueous

humor to plasma, and uveoscleral flow from the aqueous humor to the iris–ciliary body and sclera.

In all equations above, A_{drug} denotes the amount of drug in the target compartment; Surface area represents the contact area between compartments; Permeability is the permeability coefficient calculated using PK-Sim default methods; C terms denote drug concentrations in the corresponding compartments; and P_{factor} terms represent permeability modifiers accounting for ionization effects.

Drug transport within ocular tissues was modeled based on anatomical connectivity, with diffusion assumed as the sole transport mechanism. Bioavailability and unbound fraction were assumed to be unity. No active transport, metabolic clearance, melanin binding, pH-dependent permeability, or intestinal absorption was considered. Renal clearance was estimated as apparent systemic clearance using PK-Sim default settings.

2.5 | Estimation and Model Diagnostics

The model was developed following a sequential, three-step workflow. Drug-specific parameters including permeability coefficients, partition coefficients, and ocular flow rates were estimated using plasma and ocular tissue concentration data from mouse and rabbit studies. Human-specific systemic clearance was then estimated using plasma PK data obtained from a Phase I clinical trial, as human ocular tissue concentrations could not be directly measured. Finally, the model was applied to predict human ocular tissue exposure, which represents the primary translational output of this study.

Model parameter estimation was performed using both the Levenberg–Marquardt and Monte Carlo methods provided in the PK-Sim software. Initial parameter identification was conducted using the Levenberg–Marquardt method, followed by additional estimation using the Monte Carlo method to enhance model performance.

As a numerical diagnostic, a model diagnosis was conducted using the PK parameters of maximum concentration (C_{max}), average concentration (C_{avr}), and AUC. These parameters were calculated from the observed concentrations in all tissues within the dataset and were compared with the PK parameters derived from the model. The model's predictive performance was considered adequate if the geometric mean fold error (GMFE) for each tissue was 2 or less. GMFE was calculated as follows:

$$\text{GMFE} = 10^{\frac{1}{n} \sum_{i=1}^n \log_{10} \frac{\text{Predicted}_i}{\text{Observed}_i}}$$

Following the numerical diagnostics, visual diagnostics were performed to ensure that the PBPK model adequately captured the observed profiles. Study-level fit plots and goodness-of-fit assessments were conducted for visual diagnostics. Goodness of fit was evaluated for both concentration-time profiles and PK parameters across all tissues and studies by comparing the observed data with the model predictions.

To account for the unique retinal structure of mice, which includes absorption characteristics not present in rabbits or humans, such as shorter diffusion distances and a thinner sclera, the first sampling point from the mouse retina was excluded from the analysis. As species-specific physiological parameters sufficient to quantitatively characterize this difference were not available, inclusion of this timepoint distorted cross-species parameter estimates; the exclusion was therefore necessary to preserve the translational reliability of the model.

In addition, global sensitivity analysis and a study-level jackknife method were conducted to further evaluate the robustness of the model. The global sensitivity analysis identified key parameters influencing model outputs, while the jackknife approach assessed the stability of parameter estimates and predictive performance when each individual study was excluded in turn, thereby providing additional evidence for the model's structural identifiability and robustness.

2.6 | Simulation Design

The simulation was conducted to estimate an efficacious human dose based on retinal exposure. As a first step, the simulated steady-state concentrations in humans were compared to observed retinal concentrations in the CNV mouse model. Then, steady-state simulations in both species were compared. The goal was to match the human exposure PK parameters with those obtained from mouse as a reference.

The reference dose was based on 2% or 4% EB-203 administered b.i.d. in mice, which showed anti-VEGF effects. The key PK parameters were AUC and C_{avr} over the last 24 h at steady state [15, 16]. Simulated human PK profiles (multiple dosing, q.i.d.) were compared with the observed mouse PK profiles (single dosing, 2% EB-203) and simulated mouse profiles (multiple dosing, 2% and 4% EB-203, b.i.d.) to determine the therapeutic dose. For the q.i.d. regimen, dosing was administered at 0, 4, 8, and 12 h after waking (no nighttime dosing) for 9 days. Two doses (2% and 4%) were administered, whereas the b.i.d. regimen was administered every 12 h for the same duration. Human PKs are represented by boxplots (25th–75th percentiles, 1.5 IQR), whereas mouse PKs are represented by the 50th percentile line and 5th–95th percentile range.

The simulation ran 100 iterations, incorporating ocular tissue volume variability (30% CV, lognormal distribution, conservative estimate relative to reported values of approximately 10% CV in healthy adults [17]) while the other parameters remained fixed. The suitability of the therapeutic dose was evaluated by comparing the C_{max} , AUC, and C_{avr} , as well as time-series plots of simulated human tissue concentrations against simulated mouse averages. Simulation settings were summarized in Table S4.

3 | Results

3.1 | Exploratory Data Analysis

Despite anatomical differences across species, peak drug concentrations were consistently achieved within 1.5 h of administration across all ocular tissues and species examined, suggesting rapid

tissue distribution following topical instillation (Figure S2). In rabbits, the lowest drug concentrations were found in plasma, aqueous humor, and vitreous humor. Midlevel exposure was recorded in the iris-ciliary body, retina, and sclera, whereas the highest exposure was observed in the cornea, conjunctiva, and choroid. In mice, a similar pattern was observed, except that the cornea exhibited a lower exposure (Table S5). The systemic exposure was minimal, with plasma concentrations markedly lower than those observed in ocular tissues of animal models. These findings are consistent with the expected topical behavior of the drug and suggest that EB-203 is well-contained within the ocular region after administration.

3.2 | Model Parameterization

A total of 59 parameters per species (177 parameters in total) related to ocular tissue surface area, volume, and flow rates were collected for humans and rabbits, whereas some of the mouse values were extrapolated using allometric scaling (Table S6). The theoretical allometric results were visually compared to the regression-based extrapolation results (using human, rabbit, and mouse data; the regression between humans and rabbits versus mice is shown in Figure S1).

Based on the generated SMILES string from the molecular structure of EB-203, the predicted $\text{Log}P$ and $\text{Log}S$ values were -1.53 and -3.39 , respectively. To reflect formulation effects, the $\text{Log}P$ value was subsequently estimated to be 0.51 .

Other than physiological structure parameters, drug-specific model parameters, such as lipophilicity, ocular flow rates, ocular emptying rate via the nasolacrimal duct, interstitial surface area, ocular partition coefficients (between interstitial and intracellular compartments), plasma clearance, permeation multiplier parameters ("P factor"), were estimated. The specific blood flow rate to the eyes was fixed at its reference value, whereas the unbound fraction and intestinal permeability were set to 1 and 0, respectively, consistent with initial assumptions (Table 1). The chosen schemes for the eye compartments and whole PBPK are illustrated in Figure S3.

3.3 | Model Diagnostics

In numerical diagnostics, 100% of the tissue PKs by species satisfied a GMFE of less than 2 for C_{avr} and AUC, whereas 71% satisfied this criterion for C_{max} . The results are summarized in Table 2.

The study-level fit between observed and model-predicted concentrations demonstrated that the model adequately describes the concentration-time profiles across ocular tissues (Figure S4). In addition, the goodness-of-fit evaluations for both concentration-time data across tissues and PK parameters in each study confirmed that the predictions were closely aligned with observations, clustering around the unity line and mostly falling within 2-fold error boundaries (Figure 1 and Figure S5).

3.4 | Simulation Results

As described in the Methods section, simulations were performed for the administration of 2%–4% EB-203 q.i.d. to humans,

and 2%–4% EB-203 b.i.d. to mice. The results indicated that drug concentrations reached steady state without progressive day-to-day accumulation, likely attributable to the substantial overnight washout interval between dosing days, despite pronounced within-day concentration fluctuations. Additionally, within a single day, the drug concentrations appeared to reach steady-state from the third dose onward (Figure S6).

In repeated simulations, 4% b.i.d. administration in mice resulted in the highest drug concentrations, whereas 2% q.i.d. administration in humans resulted in the lowest concentrations. The 2% b.i.d. regimen in mice and 4% q.i.d. regimen in humans produced concentration intermediates between these extremes. The difference between peak and trough concentrations was more pronounced in mice.

When the simulated human PK profiles were compared with observations in mice (single dose, 2% EB-203), both the plasma and retinal AUC in humans (human retinal AUC: 200.5–396.9124.69 ng/mL·h) exceeded the observed values in mice across all doses (mouse retinal AUC: 124.69 ng/mL·h). The results for C_{avr} exhibited a similar pattern. However, the predicted C_{max} in humans (human retinal C_{max} : 17–33 ng/mL at 2% and 4% q.i.d.) was lower than that observed in mice (mouse retinal C_{max} : 73.14 ng/mL), particularly in plasma samples (Figure S7).

For the 4% q.i.d. regimen, the simulated human retinal AUC and C_{avr} (396.9 ng/mL·h and 16.5 ng/mL) fell within the median range of the simulated mouse values (mouse retinal AUC: 284.1–551.8 ng/mL·h; C_{avr} : 11.8–23.0 ng/mL at 2% and 4% b.i.d.), confirming comparable exposure between species (Table 3 and Figure 2). Consistently, in the time-series comparison of retinal drug concentration, the simulated human concentration profile encompassed the average mouse retinal concentration for each dose (Figure 3).

4 | Discussion

Exploratory data analysis indicated that ocular administration of the drug resulted in minimal systemic exposure, with the plasma AUC being approximately 1% of the highest AUC observed in ocular tissues (Table S5).

According to the goodness of fit for PK parameters by study and tissue, it was confirmed that most fold errors were well within a 2-fold error, which is commonly accepted as a satisfactory criterion in similar translational studies [20–23], our current model appears appropriate for cautious interspecies extrapolation of drug efficacy. At higher concentrations, the goodness of fit for C_{max} showed larger deviations from the unity line. For AUC, the deviations were relatively uniform across the concentration range, and all values were within the 2-fold error line, even at higher concentration levels. For C_{avr} , a trend similar to that of the AUC was observed (Figure 1).

This discrepancy in the estimated PK parameters (Table 1) may result from variations in the relationships between anatomical elements of the ocular structure [24, 25] and difficulties in the sampling process owing to its physical size, despite the core structures of the eye being conserved across species. The model

TABLE 1 | Estimated model parameters and its description.

Parameter	Initial	Estimate	Unit	Description
Lipophilicity (LogP)	−1.53	0.51	—	Lipophilicity, log units. Initial value collected from ALGOPS 2.1
Solubility (LogS)	−3.39	—	—	Solubility, log units. Value collected from ALOGPS 2.1
Uveoscleral flow rate	0	2.99×10^{-3}	mL/min/g organ	Uveoscleral flow, volumetric flow
Trabecular flow rate	0	0.02	mL/min/g organ	Trabecular flow, volumetric flow
Aqueous humor production rate	0	0.03	mL/min/g organ	Aqueous humor production rate, volumetric flow
Fraction unbound (plasma)	1 ^a	—	—	Fraction of unbound drug in plasma, no bound fraction is assumed (no data available)
Fraction interstitial	0.2	0.06	—	The marginal fraction of interstitial space available for material exchange, excluding the specified eye compartments (aqueous humor, vitreous humor, cornea, conjunctiva, sclera, iris-ciliary body, retina, and choroid). The initial value was set based on typical fraction interstitial values (ranging from 0.004 to 0.3)
Ocular transit rate (rabbit)	0.15	0.06	1/min	Tear transit rate as a first-order rate constant. Initial value set based on a nasolacrimal drainage half-life of approximately 4 min [18]
Ocular transit rate (mouse)	0.15	2.26	1/min	
Ocular transit rate (human)	0.15	0.13	1/min	
Surface correction factor	1	58.14	Unitless	Parameter used to calculate the effective surface area for material exchange between the (marginal) interstitial space and the specified eye compartments. The initial surface area was estimated assuming a spherical shape based on tissue mass, and multiplied by the Surface correction factor to derive the effective surface area
Plasma clearance (human)	0	0.33	mL/min/kg	Apparent plasma clearance in the kidney. Estimated to represent whole-body clearance
Plasma clearance (mouse)	0	31.43	mL/min/kg	
Plasma clearance (rabbit)	0	22.04	mL/min/kg	
Kp (choroid)	Default	26.70	—	Partition coefficient between intracellular/interstitial space. Default values supplied from PK-Sim standard method
Kp (retina)	Default	24.73	—	
Kp (sclera)	Default	32.31	—	
Kp (conjunctiva)	Default	2.76	—	Partition coefficient between interstitial space/plasma. Default values supplied from PK-Sim standard method
Kp (interstitial/plasma)	Default	1.89	—	
Kp (barrier/interstitial)	1	87.72	—	

(Continues)

TABLE 1 | (Continued)

Parameter	Initial	Estimate	Unit	Description
<i>P</i> factor (aqueous/iris)	1	1.12	—	Multiplier for permeability between ocular compartments. Base permeability was calculated from PK-Sim standard method
<i>P</i> factor (aqueous/vitreous)	1	99,976	—	
<i>P</i> factor (choroid/retina)	1	2177	—	
<i>P</i> factor (corn/aqueous)	1	23,945	—	
<i>P</i> factor (vitreous/retina)	1	1709	—	
<i>P</i> factor (nasolacrimal/plasma)	1	62.66	—	
<i>P</i> factor (pre-cornea/conjunctiva)	1	1.00×10^{-3}	—	
<i>P</i> factor (pre-cornea/cornea)	1	0.08	—	
<i>P</i> factor (interstitial/barrier)	1	0.17	—	
<i>P</i> factor (sclera, anterior)	1	72,949	—	
<i>P</i> factor (sclera, posterior)	1	3159	—	
Specific blood flow rate (human)	9 ^a	—	mL/min/100 g organ	Specific blood flow to eye compartment. Calculated from blood flow (mouse 0.0155 mL/min ^b , rabbit 0.504 mL/min, human 0.648 mL/min [19]) divided by volume of each species (mouse 0.025 mL, rabbit 2.6 mL, human 7.2 mL)
Specific blood flow rate (mouse)	61.9 ^a	—	mL/min/100 g organ	
Specific blood flow rate (rabbit)	19.4 ^a	—	mL/min/100 g organ	
Intestinal permeability (transcellular)	0 ^a	—	cm/min	Intestinal permeability, velocity. No oral absorption assumed

Note: To represent the pharmacokinetics after dosing in both eyes, the specific blood flow rates and dimension-related parameters for mouse and rabbit were doubled.

^aFixed value.

^bAllometric scaled parameter.

attempted to apply a uniform structural framework to all species by adjusting its permeability rate parameters, which may not have captured all the subtle differences between them. For this reason, the first mouse retinal point, which is an atypically high peak (968.55 ng/mL at 0.083 h, 52.12 ng/mL at 0.25 h in geometric mean; approximately 20 times larger than the adjacent sampling point), was excluded when applying this common structure; to evaluate the exclusion's influence, the full analysis with the point retained is reported in Data S2.

The contribution of each absorption route was estimated from the fractional AUC in downstream compartments receiving drug via each pathway (Figure S8), suggesting that the nasolacrimal duct contributed 38.77%, the conjunctiva 26.68%, and the cornea 34.53% to overall absorption. This pattern aligns well with the drug log*P* estimate, a measure of drug's lipophilicity, which plays a critical role in drug absorption together with solubility [26–32]. In other words, higher lipid solubility favors corneal absorption, whereas lower lipid solubility (higher water solubility) favors conjunctival absorption. The relatively balanced distribution across routes corresponds to the drug's intermediate log*P* value.

Global sensitivity analysis results across compartments (plasma, ocular plasma, and retina) revealed a consistent pattern in which the most influential parameters were related to the physiological transport dynamics rather than to isolated structural attributes (Figure S9). In particular, nasolacrimal duct-associated parameters, such as the ocular transit rate

permeation multiplicate factor (*P* factor), consistently showed a strong influence on systemic and ocular exposure. These results highlight the importance of drainage-mediated elimination pathways in shaping the overall pharmacokinetic profile following topical administration.

Retinal drug exposure was mainly influenced by parameters related to permeability into the eye, reflecting the combined effects of trans-ocular transport and local diffusional properties in determining posterior segment distribution. Permeability across the precorneal and corneal regions also contributed indirectly, suggesting that ocular surface dynamics may affect drug concentrations in deeper tissues. Parameters such as lipophilicity and hematocrit, which function as scaling factors by multiplying exposure levels, showed similar influence patterns. Rather than a single dominant factor driving the model, multiple parameters contributed in a balanced manner, indicating that the exposure was shaped by an integrated set of physiological and physico-chemical properties.

Additionally, parameter identifiability within the model was supported by the nature of the data and the parameter roles: single- and multiple-dose data primarily informed the estimation of drug clearance, while ocular transit rates (drainage) and tissue flows governed elimination rates within ocular tissues. Permeability and partition coefficients mainly determined the concentration gradients and PK profile differences across ocular compartments, acting in conjunction with the estimated values

of these parameters. This was further supported by jackknife analysis (Figure S10), which demonstrated that parameter estimates did not exhibit extreme values, indicating model robustness and generalizability within the dataset.

Several limitations of this study should be acknowledged. The model was constructed assuming passive diffusion as the sole drug transport mechanism, with f_u set to 1 and no melanin binding, active transport, or metabolic clearance incorporated. While this approach is supported by the physicochemical and pharmacokinetic characteristics of EB-203, the absence of in vitro data on these processes means that uncertainty remains regarding

whether active mechanisms, if present, differ between species in ways that could affect the accuracy of human ocular exposure predictions. Additionally, interspecies differences in ocular barrier properties and potential differences in target engagement were not explicitly parameterized in the model, which represents an additional source of uncertainty in mouse-to-human scaling that should be considered when interpreting the predicted human retinal exposures. Because the model estimates many parameters from a single compound, some interface parameters are only weakly constrained, and those converging toward their bounds indicate that the data place those pathways at one extreme rather than at a specific value. Extending the framework to ophthalmic agents with different ADME properties would constrain these parameters and improve its generalizability.

TABLE 2 | Geometric mean fold error of tissue by species.

Species	Compartment	GMFE		
		$C_{\max}$	C_{avr}	AUC
Human	Plasma	1.37	1.30	1.30
Mouse	Choroid	4.41	1.80	1.80
	Cornea	1.72	1.51	1.51
	Plasma	5.39	1.90	1.90
	Retina	1.38	1.25	1.25
Rabbit	Aqueous humor	2.04	1.17	1.17
	Choroid	1.31	1.08	1.08
	Conjunctiva	1.73	1.61	1.61
	Cornea	1.66	1.43	1.43
	Iris ciliary body	1.97	1.12	1.12
	Plasma	1.20	1.30	1.30
	Retina	2.82	1.25	1.25
	Sclera	1.18	1.49	1.49
	Vitreous humor	1.25	1.23	1.23

Additionally, a formal PK-PD analysis directly linking retinal drug exposure to efficacy endpoints was not performed in this study; the translational inference drawn here relies on the assumption that AUC and C_{avr} are the primary determinants of anti-VEGF efficacy for EB-203, which warrants direct investigation in future work.

During parameter estimation, significant interspecies differences in plasma clearance were observed, with human clearance being substantially lower than that in mice and rabbits. This discrepancy may be attributed to a lack of consideration for bioavailability, which is influenced by the dimensional characteristics of the cul-de-sac of the eye [33]. Because no intravenous injection data were available to determine bioavailability, this factor was not explicitly incorporated into the model. When comparing the administered drug volumes (50 μL for humans/rabbits and 5 μL for mice per eye) to the cul-de-sac capacities of each species, it was evident that the administered volumes exceeded the cul-de-sac capacity in animals. Consequently, drug loss was likely; however, the bioavailability could not be directly quantified owing to insufficient data. Instead, the high plasma concentration observed in the animals was interpreted as apparent clearance [34], which adequately explained the PK profile. The discrepancy in the model fit at higher doses, as revealed

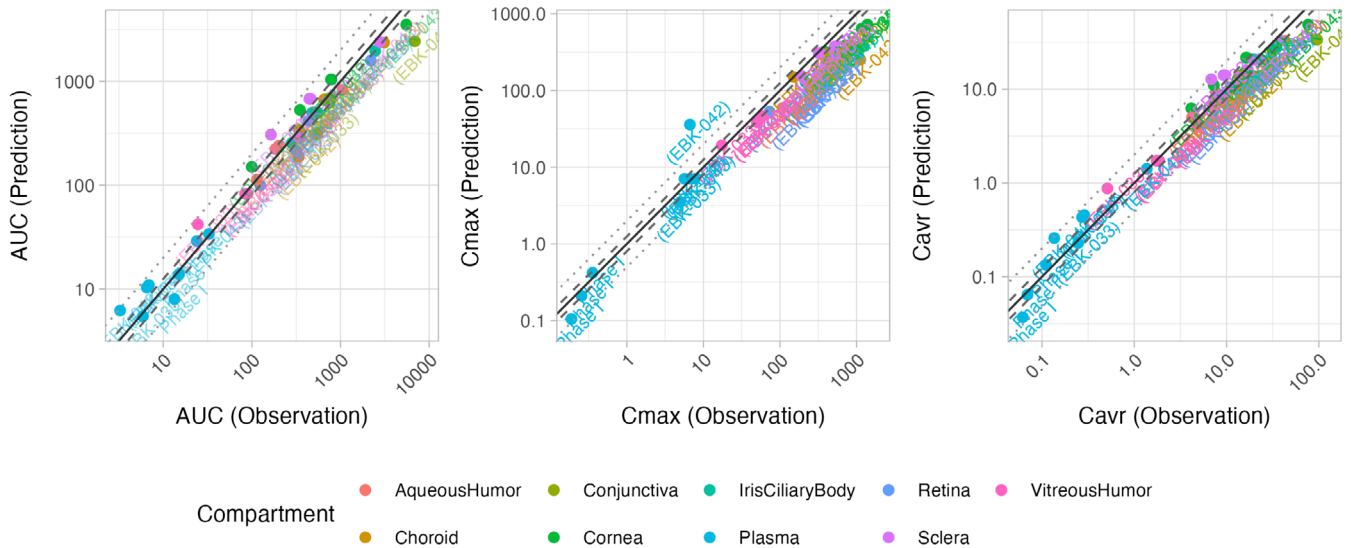


FIGURE 1 | Goodness of fit for AUC (left), $C_{\max}$ (middle), and C_{avr} (right) between observation and prediction from tissues and studies. The first mouse retinal sampling point was excluded from both the estimation and the plot.

TABLE 3 | Predicted steady-state $C_{\max}$, C_{avr} , and AUC of retinal concentration.

Dose	Species	$C_{\max}$ (ng/mL, SD)	C_{avr} (ng/mL, SD)	AUC (ng·h/mL, SD)
2%	Human	17 (1.5)	8.4 (0.4)	200.5 (9.6)
4%		33 (3.2)	16.5 (1)	396.9 (23.3)
2%	Mouse	58.3 (8.7)	11.8 (1.6)	284.1 (38.5)
4%		115.2 (16.4)	23 (2.9)	551.8 (68.7)

Note: Mean (SD) of 100 iterations for 9 days of simulation. PK parameters were derived by NCA from simulated steady-state profiles (last 24-h), human: q.i.d. mouse: b.i.d.
Abbreviations: AUC, area under the concentration-time curve; b.i.d., twice daily; C_{avr} , average concentration; $C_{\max}$, maximum concentration; q.i.d., four times daily; SD, standard deviation.

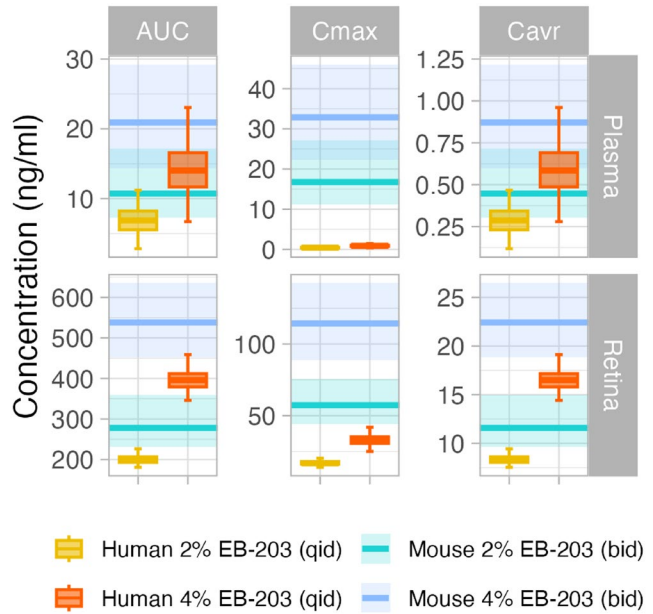


FIGURE 2 | Boxplot of simulated human PK parameters (q.i.d. dosing) and simulated mouse (b.i.d. dosing) values as criteria (line: Median and area: 5th–95th percentile).

by the goodness-of-fit assessment, may also be attributed to the omission of drug loss in the analysis.

The observed high clearance, combined with differences in the ocular emptying rates that transfer the drug to the stomach, also contributed to apparent variations in clearance among animal species. While metabolism, protein binding, melanin binding [35], or transporter-mediated processes and their potential kinetics may have influenced the behavior of the drug, they could not be incorporated into the analysis because of a lack of identifiable data.

The PBPK model was developed based on various established ocular modeling principles [36, 37], incorporating a refined structure by separating the interstitial fluid compartments inside and outside the barrier. This approach allowed for a more precise representation of ocular tissue interactions, particularly by leveraging sampling data from multiple tissues in rabbits, which have an ocular structure similar to that of humans.

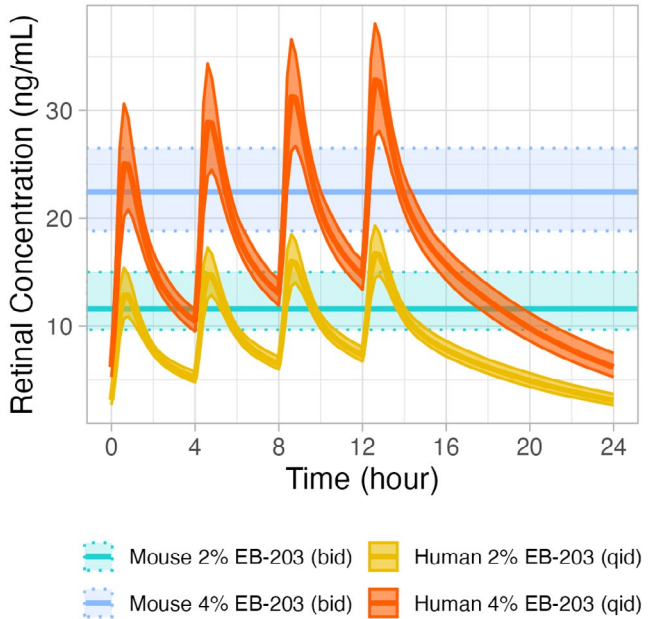


FIGURE 3 | Time series comparison between human retinal concentration profile in q.i.d. dosing (line: Median, area: 5th–95th percentile) and mouse average retinal concentration in b.i.d. dosing (horizontal line: Median, area: 5th–95th percentile).

Consequently, this model could be effectively utilized to evaluate other ophthalmic formulations.

When comparing single-dose mouse observations with multiple-dose human simulations (Figure S7), drug exposure in humans is expected to exceed the effective levels established in mice. Simulations indicated that a dosing regimen of 4% EB-203 q.i.d. in humans would achieve exposure comparable to that of 2% and 4% EB-203 administered b.i.d. in mice, satisfying the minimum exposure-related conditions based on AUC and C_{avr} . Therefore, the optimal efficacious dose for humans is likely between 2% and 4% EB-203 q.i.d., with 4% q.i.d. confidently providing sufficient therapeutic exposure.

In summary, the ocular model demonstrated adequate performance in predicting human ocular drug exposure, providing a mechanistic basis for translational dose selection. Given the refined differentiation of the key anatomical compartments, the developed model is expected to be beneficial for evaluating other types of ophthalmic formulations.

5 | Conclusion

This study successfully established a physiologically based pharmacokinetic (PBPK) model for EB-203, enabling the quantitative prediction of human ocular tissue exposure based on pre-clinical data. The model effectively integrated anatomical and physiological information across species and demonstrated good predictive performance, with most key pharmacokinetic parameters falling within an acceptable range of prediction error.

Simulation results suggest that a 4% q.i.d. dosing regimen of EB-203 in humans can achieve retinal drug exposure levels comparable to those associated with therapeutic efficacy in the

CNV mouse model. These findings indicate that EB-203 has the potential to exert a therapeutic effect in humans, supporting its continued clinical development.

Importantly, this study highlights the practical utility of ocular PBPK modeling as a translational tool to bridge preclinical and clinical pharmacokinetics in the absence of direct human tissue measurements. The modeling framework developed here can be adapted and extended to other ophthalmic drug candidates, providing a mechanistic foundation for rational dose selection and enhancing the efficiency of ophthalmic drug development.

Author Contributions

W.J. wrote the manuscript; W.J., S.L. J.Y. J.-w.C. and H.Y. designed the research; W.J., J.Y., J.J.L., Y.C. performed the research; W.J. and K.A.M. analyzed the data.

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Conflicts of Interest

The authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** cts70717-sup-0001-DataS1.zip. **Data S2:** Sensitivity to early time point. **Figure S1:** Log-log plot and regression results of allometric scaled parameters. Solid line: regression line derived from allometric scaling between human and rabbit data. Dashed line: line connecting individual data points, provided as a visual guide for trend direction. **Figure S2:** Distribution of NCA (non-compartmental analysis) parameters (C_{max} , AUClast, T_{max}) by tissue type across the included studies. Point: Point estimate of NCA parameter in each study, Bar: Median value. **Figure S3:** Scheme of eye compartment and whole-PBPK model. **Figure S4:** Study-level fit by tissue. **Figure S5:** Goodness of fit for concentration-time point between observation and prediction by tissues. **Figure S6:** Simulated plasma and retinal concentrations were generated for both humans and mice under each dosing scenario. Dosing frequency were q.i.d. for human and b.i.d. for mouse. **Figure S7:** PK parameter comparison with simulated human (q.i.d. dosing, bar) and observed mouse values (2% EB-203 single dosed, geometric mean, blue horizontal line) in plasma and retina. The first mouse retinal sampling point was excluded from both the estimation and the plot. **Figure S8:** Simulated drug amount in four consecutive dosing in different tissues (plasma: plasma in eye compartment which receives the drug amount directly from nasolacrimal duct). **Figure S9:** Sensitivity analysis was conducted for each organ (eye, venous blood) and compartment (plasma, retina). All parameters included in the model were evaluated, and the top 10 parameters ranked by absolute sensitivity were displayed. A sensitivity value of –1.0 indicates that a 10% increase in the parameter results in a 10% decrease in the PK parameter value, while a sensitivity of +0.5 indicates a 5% increase in the PK parameter value. **Figure S10:** Distribution of parameter estimates obtained using the jackknife method, where each dataset was sequentially excluded from the estimation. Parameter estimates were transformed into z-scores to visualize their standardized distribution across the jackknife replicates. Red circles: final parameter estimates, Black circles: sampled estimates in each jackknife replicates. Parameters that were fixed or remained unchanged were not displayed. **Table S1:** Key design features in collected studies. **Table S2:** Bioanalytical method used in collected studies. **Table S3:** Compartment configuration. **Table S4:** Settings of simulation scenario in comparison of simulations and observation. **Table S5:** Included datasets and its Tissue AUC (noncompartmental analysis) results. **Table S6:** Collected/Generated physiologic values for ocular compartments.